

# Set 7: Customer Demographics Analysis

DSA – Data Handling and Visualization | Lab Experiment

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## Aim

To visualise the age, gender and income profile of customers using a bar chart, a pie chart and a labelled table, and to combine them into an interactive Tableau dashboard.

## Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

## Dataset

| Customer ID | Age | Gender | Income (in \$) |
|-------------|-----|--------|----------------|
| 1           | 28  | Female | 50,000         |
| 2           | 35  | Male   | 60,000         |
| 3           | 42  | Female | 75,000         |

## Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
customers <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
write.csv(customers, "Customer_Demographics.csv", row.names = FALSE) # for Tableau
```

## Q1. Bar Chart – Distribution of Customer Ages

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

### R Program:

```
# Q1. Bar chart - distribution of customer ages
print(customers[, c("Customer_ID", "Age")])
cat("Mean age:", mean(customers$Age), "| Range:", min(customers$Age), "-",
    max(customers$Age), "\n")

p_age <- ggplot(customers, aes(x = factor(Customer_ID), y = Age, fill = Gender)) +
  geom_col(width = 0.55) +
  geom_text(aes(label = paste(Age, "yrs")), vjust = -0.5, size = 4) +
  scale_fill_manual(values = c(Female = "#CC79A7", Male = "#0072B2")) +
  scale_y_continuous(limits = c(0, 50), breaks = seq(0, 50, 10)) +
  labs(title = "Distribution of Customer Ages",
       x = "Customer ID", y = "Age (years)", fill = "Gender") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_age)
ggsave("s7_q1_age_bar.png", p_age, width = 7, height = 4.5, dpi = 150, bg = "white")
```

### Console Output:

```
Customer_ID Age
1          1  28
2          2  35
3          3  42
Mean age: 35 | Range: 28 - 42
```

**Output:**

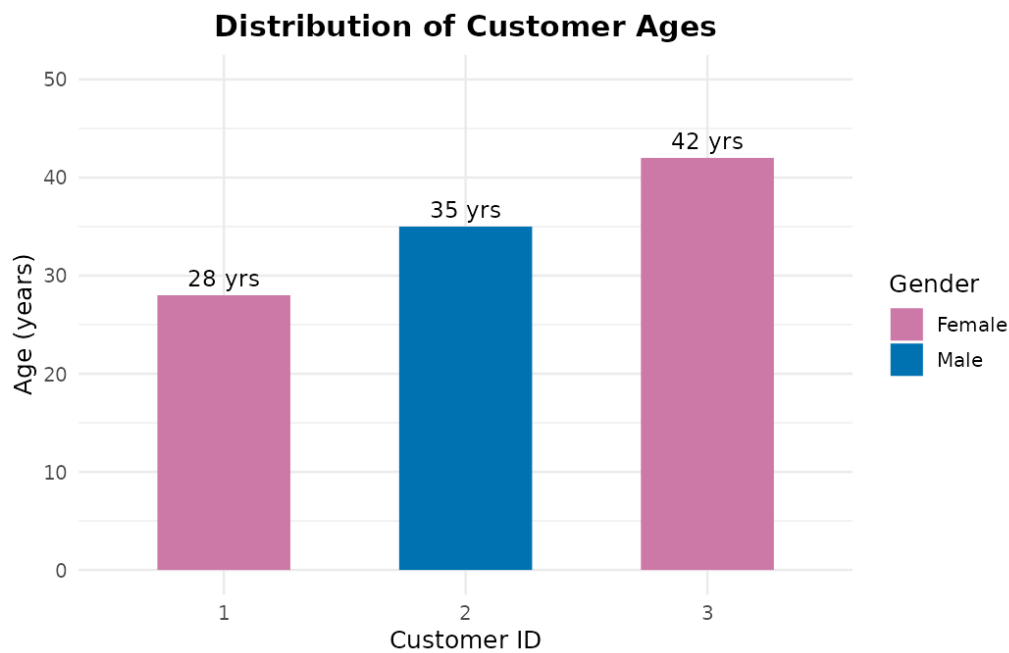


Figure 1: Distribution of customer ages

**Inference:**

Customer ages range from 28 to 42 years with a mean of 35. Ages rise evenly by 7 years from Customer 1 to Customer 3, so the customer base is spread across young and middle-aged adults, with no dominant age group.

## Q2. Pie Chart – Customers by Gender

Generate a pie chart to represent the distribution of customers by gender.

**R Program:**

```
# Q2. Pie chart - customers by gender
gender_count <- as.data.frame(table(customers$Gender))
names(gender_count) <- c("Gender", "Count")
gender_count$Percent <- round(100 * gender_count$Count / sum(gender_count$Count), 1)
print(gender_count)

p_pie <- ggplot(gender_count, aes(x = "", y = Count, fill = Gender)) +
  geom_col(width = 1, colour = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = paste0(Gender, "\n", Count, " (", Percent, "%)")),
    position = position_stack(vjust = 0.5), colour = "white", size = 4.5) +
  scale_fill_manual(values = c(Female = "#CC79A7", Male = "#0072B2")) +
  labs(title = "Distribution of Customers by Gender", fill = "Gender") +
  theme_void(base_size = 12) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_pie)
ggsave("s7_q2_gender_pie.png", p_pie, width = 5.5, height = 4.5, dpi = 150, bg = "white")
```

**Console Output:**

```
  Gender Count Percent
1 Female     2    66.7
2  Male     1    33.3
```

Output:

## Distribution of Customers by Gender

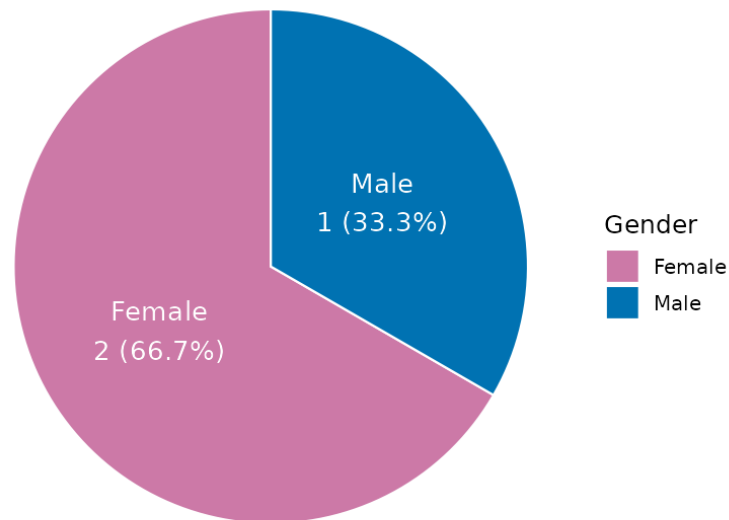


Figure 2: Customers by gender

### Inference:

Two of the three customers (66.7%) are female and one (33.3%) is male, so the customer base is female-majority.

## Q3. Table – Demographic Information for Each Customer

Build a table to show the demographic information for each customer. Label the table elements.

### R Program:

```
# Q3. Labelled table of customer demographics
tbl <- customers
tbl$Income <- paste0("$", format(tbl$Income, big.mark = ","))
names(tbl) <- c("Customer ID", "Age (years)", "Gender", "Income (in $)")
print(tbl, row.names = FALSE)

tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Customer Demographic Information",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s7_q3_table.png", width = 900, height = 330, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

### Console Output:

| Customer ID | Age (years) | Gender | Income (in \$) |
|-------------|-------------|--------|----------------|
| 1           | 28          | Female | \$50,000       |
| 2           | 35          | Male   | \$60,000       |
| 3           | 42          | Female | \$75,000       |

Output:

### Customer Demographic Information

| Customer ID | Age (years) | Gender | Income (in \$) |
|-------------|-------------|--------|----------------|
| 1           | 28          | Female | \$50,000       |
| 2           | 35          | Male   | \$60,000       |
| 3           | 42          | Female | \$75,000       |

Figure 3: Labelled customer demographics table

#### Inference:

The table shows every attribute with labelled column headers and units. Income rises with age: Customer 3 (42) earns the most at \$75,000 and Customer 1 (28) the least at \$50,000.

## Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

#### R Program:

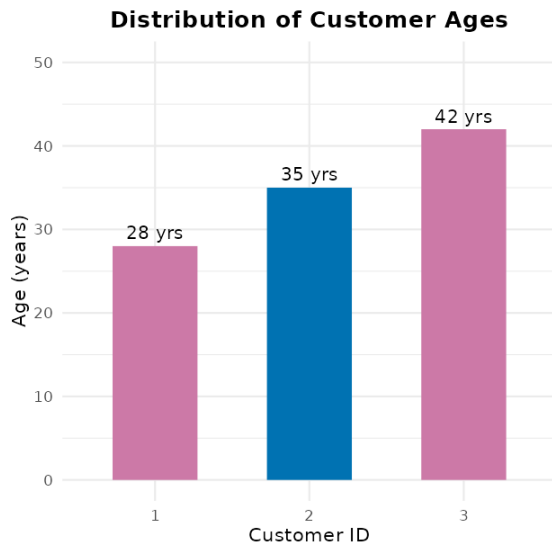
```
# Q4. Dashboard layout preview (Tableau dashboard is built from Customer_Demographics.csv)
library(patchwork)
dash <- (p_age | p_pie) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Customer Demographics Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s7_dashboard.png", dash, width = 11, height = 7, dpi = 130, bg = "white")
```

#### Procedure in Tableau:

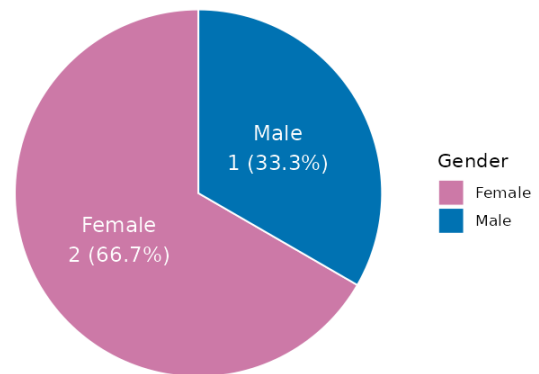
- 1 Open Tableau → **Connect** → **Text file** → select **Customer\_Demographics.csv** (written by the R program). In the Data Source tab, change **Customer ID** to a **Dimension (String)**.
- 2 **Sheet 1 – "Age Distribution"**: drag Customer ID to **Columns** and Age to **Rows**. Drag Gender to **Color** and Age to **Label**. Title: "Distribution of Customer Ages"; axes: "Customer ID", "Age (years)".
- 3 **Sheet 2 – "Gender Split"**: set the Marks type to **Pie**. Drag Gender to **Color** and **CNT(Customer ID)** to **Angle**. Drag Gender and CNT(Customer ID) to Label, and add a **Quick Table Calculation** → **Percent of Total**.
- 4 **Sheet 3 – "Customer Table"**: drag Customer ID, Gender, Age and Income to **Rows** as discrete fields (or Measure Names / Measure Values to Text). Format Income as Currency (\$). Rename the headers.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: Age Distribution (top-left), Gender Split (top-right), Customer Table (bottom). Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Customer\_Demographics) so one filter controls every view.
- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.

### Dashboard Layout:

## Customer Demographics Dashboard



## Distribution of Customers by Gender



| Customer ID | Age (years) | Gender | Income (in \$) |
|-------------|-------------|--------|----------------|
| 1           | 28          | Female | \$50,000       |
| 2           | 35          | Male   | \$60,000       |
| 3           | 42          | Female | \$75,000       |

Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

### Inference:

Clicking the "Female" slice filters the bar chart and table to Customers 1 and 3. Selecting a bar shows that customer's full record, so all three views can be explored together.

## Result

The customer demographics were visualised with a labelled bar chart, pie chart and table in R, and combined into an interactive Tableau dashboard. The visualizations show a female-majority customer base (66.7%) aged 28–42, with income increasing with age.